BURN CLASSIFICATION & MANAGEMENT

Clinical Guideline for AI Model Training Data — Version 2

Sources: WHO Burn Management | WHO White Phosphorus Fact Sheet | MSF Clinical Guidelines | DermNet NZ | MSD Manual | ICRC War Surgery | CDC/NIOSH | PMC / StatPearls

For use in low-resource and conflict settings

IMPORTANT: All content in this document is extracted strictly from published medical sources listed in the References section. No information has been added beyond what is stated in those sources. This document is intended as structured training data for AI models in clinical burn assessment.

1. BURN DEPTH CLASSIFICATION SYSTEM

Burns are classified based on depth of injury into the skin layers. The modern clinical approach uses depth/thickness terminology, which maps to the traditional degree system (WHO Burn Management; DermNet NZ; MSD Manual):

Depth Classification	Traditional Term	Skin Layers Involved
Superficial	First degree	Epidermis only
Superficial partial-thickness	Second degree (superficial)	Epidermis + papillary dermis
Deep partial-thickness	Second degree (deep)	Epidermis + reticular dermis
Full-thickness	Third degree	Epidermis, full dermis + subcutaneous fat

Note: It is common to find all types of burns within the same wound, and depth may change with time, especially if infection occurs. Estimating depth is difficult and burns are often underestimated on initial examination. Differentiation between superficial and deep partial-thickness may only be possible after day 8–10. (WHO Burn Management; DermNet NZ; MSF)

2. BURN LEVELS: APPEARANCE, SENSATION & INITIAL MANAGEMENT

2.1 SUPERFICIAL BURN (First Degree)

Appearance	Sensation / Pain	Blanching
Red, warm, dry. No blisters. Area turns white when touched.	Painful and warm to touch.	Blanches with light pressure.

Sources: WHO Burn Management; DermNet NZ; MSD Manual; Stanford Health Care

Initial Management (Low/No-Resource Setting):

- Stop the burning — remove clothing and stop all heat exposure.
- Cool with cool running water for approximately 20 minutes.
- Remove rings, watches, and constricting items early before swelling develops.

• Cover with a clean, non-adherent dressing or clean cloth.
• Do NOT apply ice directly, oils, toothpaste, butter, or any home remedies.

Sources: WHO Burn Management; WHO Fact Sheet on Burns; MSF Clinical Guidelines

2.2 SUPERFICIAL PARTIAL-THICKNESS BURN (Second Degree — Superficial)

Appearance	Sensation / Pain	Blanching
Red or pink, moist, blisters present (intact). Swollen.	Very painful — painful to temperature and air.	Blanches with pressure.

Sources: WHO Burn Management; DermNet NZ; MSD Manual; MSF Clinical Guidelines

Initial Management (Low/No-Resource Setting):

• Apply the same initial cooling as for superficial burns.
• Do NOT open or burst blisters — they protect underlying tissue from infection.
• Cover with a clean, non-adherent dressing.
• Monitor for pain, swelling, and signs of infection (spreading redness, discharge, odor).
• Healing usually occurs within 3 weeks without surgical intervention.

Sources: WHO Burn Management; MSF Clinical Guidelines; DermNet NZ

2.3 DEEP PARTIAL-THICKNESS BURN (Second Degree — Deep)

Appearance	Sensation / Pain	Blanching
Variable — mottled red/waxy white. Surface moist or dry. Blisters may be broken. Hair usually absent.	Pain reduced vs. superficial partial-thickness — nerve endings partially damaged. Perceives deep pressure only.	Minimal or no blanching. Sluggish capillary response.

Sources: WHO Burn Management; DermNet NZ; StatPearls/NCBI Burn Classification; MSF Clinical Guidelines

Initial Management (Low/No-Resource / Conflict Setting):

• Apply a clean, dry covering — do NOT apply folk remedies or popular ointments.
• Treat as higher-risk injury: higher risk of delayed healing, scarring, and infection.
• Refer for specialist care as soon as possible — may require skin grafting if no spontaneous healing within 3–5 weeks.
• This burn type may be difficult to distinguish from superficial partial-thickness on day 1; clearer after day 8–10.

Sources: WHO Burn Management; MSF Clinical Guidelines; StatPearls/NCBI Burn Classification

2.4 FULL-THICKNESS BURN (Third Degree)

Appearance	Sensation / Pain	Blanching
Pearly white, leathery gray, brown, or charred black. Dry, stiff, inelastic. Hair follicles and sweat glands destroyed. Fat/muscle/bone may be involved in severe cases.	Little or no pain centrally — nerve endings destroyed. Only deep pressure perceived. Surrounding tissue may remain painful.	Does NOT blanch. No blanching due to compromised blood supply.

Sources: WHO Burn Management; DermNet NZ; MSD Manual; MSF Clinical Guidelines; StatPearls/NCBI; Stanford Health Care

Initial Management (Low/No-Resource / Conflict Setting):

• Cover with a clean, dry dressing — do NOT attempt vigorous cleaning.
• Do NOT attempt to remove clothing or materials adherent to the burn.
• Maintain the patient's general body temperature — use clean sheets or survival blanket if available.
• SEVERE burn — requires specialist care as soon as possible. Does not heal without surgery.

• Requires skin grafting after wound excision or appearance of healthy granulation tissue.

Sources: WHO Burn Management; MSF Clinical Guidelines; DermNet NZ; MSD Manual

3. COMPARATIVE SUMMARY TABLE

Parameter	Superficial	Superficial Partial-Thick.	Deep Partial-Thick.	Full-Thickness
Traditional term	1st degree	2nd degree (superficial)	2nd degree (deep)	3rd degree
Layers involved	Epidermis only	Epidermis + papillary dermis	Epidermis + reticular dermis	Epidermis + full dermis + subcutaneous fat
Color	Red, warm	Red / pink	Mottled red-white / waxy	White, grey, brown, or charred black
Surface	Dry	Moist / weeping	Moist or dry	Dry, stiff, leathery
Blisters	None	Yes (intact)	May be present or broken	None
Blanching	Yes	Yes	Minimal / sluggish	No
Pain level	Painful	Very painful	Reduced (pressure only)	Little or none
Hair follicles	Intact	Intact	Usually absent	Destroyed
Healing without surgery	Few days	Up to 3 weeks	3–5 weeks (high scar risk)	Does not heal without surgery

Sources: WHO Burn Management; DermNet NZ; MSF Clinical Guidelines; MSD Manual; StatPearls/NCBI Burn Classification

4. FLUID RESUSCITATION IN BURN PATIENTS (EXPANDED)

Fluid resuscitation is a critical intervention in the first 24 hours after a major burn injury. The inflammatory response triggered by a major burn increases capillary permeability, causing intravascular fluid, plasma proteins, and electrolytes to shift into the interstitial space. This fluid shift leads to hypoperfusion, decreased cardiac output, and if untreated, organ ischemia. This process begins immediately after the burn and peaks 8–12 hours later. (StatPearls — Burn Fluid Resuscitation; Parkland Formula — StatPearls/NCBI)

4.1 Indications for IV Fluid Resuscitation

Patient Group	Threshold for IV Fluid Resuscitation
Adults	Burns greater than 15–20% TBSA (deep partial-thickness or full-thickness)
Children	Burns greater than 10% TBSA
Elderly	Burns greater than 10% TBSA or any significant burn with comorbidities
All groups — regardless of TBSA	Any full-thickness burn; electrical burn; inhalation injury; associated trauma

Note: First-degree (superficial) burns are NOT included in TBSA calculation and do NOT warrant IV fluid resuscitation. Only second- and third-degree burns count toward TBSA. (StatPearls — Parkland Formula; MSD Manual; WHO Burn Management)

4.2 Fluid of Choice

• Lactated Ringer's solution (also called Ringer's Lactate or Hartmann's solution) is the preferred crystalloid for initial resuscitation across all age groups. (StatPearls — Burn Fluid Resuscitation; MSF Clinical Guidelines)

• Crystalloids are preferred over hypotonic fluids, which are avoided as they can exacerbate edema. (StatPearls — Burn Fluid Resuscitation)
• For pediatric patients: Lactated Ringer's is used for resuscitation; infants may additionally require dextrose supplementation due to limited glycogen stores. (StatPearls — Burn Fluid Resuscitation)
• MSF protocol: Ringer's Lactate (RL) 20 ml/kg during the first hour, even if the patient is stable. (MSF Clinical Guidelines)
• Avoid 0.9% normal saline as the sole resuscitation fluid — risk of hyperchloremic acidosis. (Modified Parkland Formula guidance; StatPearls)

4.3 The Parkland Formula — Adults

The Parkland formula is the most widely used burn resuscitation formula. It was developed by Dr. Charles Baxter at Parkland Memorial Hospital in 1968. (StatPearls — Parkland Formula; Wikipedia — Parkland Formula)

PARKLAND FORMULA — ADULTS

Total Lactated Ringer's for first 24 hours = **4 mL × Body Weight (kg) × % TBSA burned**

Example: Adult weighing 80 kg with 20% TBSA burn:
4 × 80 × 20 = **6,400 mL over 24 hours**

Half (50%) given in first 8 hours from time of burn.
Remaining half given over next 16 hours.

Sources: StatPearls — Parkland Formula (NCBI); StatPearls — Burn Fluid Resuscitation; Wikipedia — Parkland Formula; Taylor & Francis — Parkland Formula

Important notes on timing:

• Time starts from the time of burn — NOT from the time of arrival at the healthcare facility.
• If the patient arrives 2 hours after the burn, the remaining first 8-hour volume must be delivered in the remaining 6 hours.
• Subtract any fluids already given before arrival when calculating remaining volumes.
• The formula is a starting estimate only — the infusion rate must be adjusted continuously based on urine output and clinical response.

4.4 Pediatric Fluid Resuscitation

PARKLAND FORMULA — CHILDREN

Resuscitation fluid = **3 mL × Body Weight (kg) × % TBSA burned** (Lactated Ringer's)
PLUS maintenance fluids calculated by the 4-2-1 rule (Holliday-Segar method)

Half of resuscitation volume given in first 8 hours from time of burn.
Remaining half given over next 16 hours.

Maintenance fluids (4-2-1 rule):
4 mL/kg/hr for first 10 kg + 2 mL/kg/hr for next 10 kg + 1 mL/kg/hr for each kg above 20 kg

Sources: StatPearls — Burn Fluid Resuscitation; Osmosis — Parkland Formula; Modified Parkland Formula guidance

• Children have a higher body surface area-to-weight ratio and require proportionally more fluid per kg than adults.

• Children are more prone to hypoglycemia — maintenance fluids with dextrose may be needed for infants.
• Regular assessment of pulse rate, capillary refill, and blood pressure is essential in children as signs of cardiovascular collapse may appear late.
• Target urine output in children: 1.0 mL/kg/hour (infants: 2 mL/kg/hour). (Taylor & Francis — Parkland Formula)

4.5 Monitoring Fluid Resuscitation — Target Urine Output

Patient Group	Target Urine Output	Clinical Significance
Adults	0.5–1.0 mL/kg/hour (30–50 mL/hour in a 70-kg adult)	Primary endpoint for adequacy of resuscitation
Children (<30 kg)	1.0 mL/kg/hour	Adjust fluid rate to maintain target
Infants	2.0 mL/kg/hour	Monitor closely; higher risk of hypoglycemia
Electrical burns	1.0–2.0 mL/kg/hour	Higher target to protect kidneys from myoglobinuria

Sources: StatPearls — Parkland Formula; Taylor & Francis — Parkland Formula; ScienceDirect — Parkland Formula overview; MSF Clinical Guidelines

Adjusting fluid rate based on urine output:

• If urine output is below target: increase fluid infusion rate by 20–25%.
• If urine output is above target and hemodynamically stable: reduce fluid infusion rate by 20–25%.
• A Foley catheter is mandatory for burns involving more than 15% TBSA and for all electrical burns as urine output is the single best indicator of resuscitation adequacy. (ScienceDirect; WHO Burn Management)
• A urinary catheter should also be inserted for burns of the perineum/genitalia. (MSF Clinical Guidelines)
• Other variables to monitor alongside urine output: heart rate, blood pressure, mental status, serum lactate, and base deficit. (StatPearls — Burn Fluid Resuscitation)

4.6 Special Situations Requiring Higher Fluid Volumes

Condition	Effect on Fluid Requirements
Inhalation injury	Consistently requires more fluid — increase volumes by 50% (3 mL/kg × %BSA for first 8 hours). (MSF Clinical Guidelines)
Electrical burns	Deeper tissue damage than externally apparent — often requires higher end of formula range or above. (MSF Clinical Guidelines; StatPearls)
Delayed presentation	May require more aggressive initial resuscitation to compensate for under-resuscitation in the first hours. (StatPearls)
Burns > 50% TBSA	Limit fluid calculation to 50% TBSA to avoid over-resuscitation. (MSF Clinical Guidelines)
Over-resuscitation ('fluid creep')	Can cause pulmonary edema, compartment syndrome, and multi-organ dysfunction. Avoid giving volumes exceeding Ivy index (250 mL/kg/day). (ScienceDirect; StatPearls)

Sources: MSF Clinical Guidelines; StatPearls — Parkland Formula & Burn Fluid Resuscitation; ScienceDirect — Parkland Formula

5. SPECIAL BURN TYPES: CHEMICAL & ELECTRICAL

5.1 Chemical Burns

• Chemical burns are divided into acid or alkali burns. Alkali burns tend to be more severe, causing deeper penetration into the skin through liquefaction necrosis. Acid burns penetrate less because they cause coagulation necrosis. (StatPearls — Burn Evaluation; NCBI)
• In chemical burns: flush with large volumes of water for 15 to 30 minutes, avoiding contamination of healthy skin. (WHO Fact Sheet on Burns; MSF Clinical Guidelines)

- Do NOT attempt to neutralize the chemical agent — this can worsen the injury. (MSF Clinical Guidelines)
- Remove contaminated clothing during or after irrigation, with appropriate protection for the rescuer.

Sources: WHO Fact Sheet on Burns; MSF Clinical Guidelines; StatPearls/NCBI

5.2 Electrical Burns

- Electrical burns can be deceptive: small entry and exit wounds may mask extensive internal organ injury or associated traumatic injuries. (StatPearls — Burn Evaluation)
- Clinical manifestations vary with type of current. Look for arrhythmia, rhabdomyolysis, and neurological disorders. (MSF Clinical Guidelines)
- Always treat as high risk even when external signs appear minor. (MSF Clinical Guidelines)
- Do NOT start first aid before ensuring your own safety — switch off the electrical current first. (WHO Fact Sheet on Burns)
- Increase fluid replacement volumes by 50% vs. standard calculations in the event of electrical burns due to deeper tissue damage. (MSF Clinical Guidelines)
- Monitor for myoglobinuria (dark urine) — target urine output of 1.0–2.0 mL/kg/hour to protect kidneys. (MSF Clinical Guidelines)

Sources: WHO Fact Sheet on Burns; MSF Clinical Guidelines; StatPearls/NCBI

6. WHITE PHOSPHORUS (WP) BURNS — CONFLICT SETTINGS

6.1 Background and Mechanism

- White phosphorus (WP) is a white to yellow waxy solid with a garlic-like odour. It ignites spontaneously in air at temperatures above 30–34°C and continues to burn until it is fully oxidized or deprived of oxygen. Combustion temperatures exceed 800°C. (WHO White Phosphorus Fact Sheet; PMC — WP Pathophysiology 2025)
- WP burns result from a combined thermal and chemical injury: (1) the heat of chemical combustion, and (2) the corrosive action of phosphoric acid formed as phosphorus reacts with oxygen and water in tissues. (PMC — WP Civilian Burns 2024; Burns Journal — Management of WP Burns)
- WP sticks to skin and clothing, can penetrate deep into tissues even through bone, and will continue to burn as long as it is exposed to oxygen. (WHO White Phosphorus Fact Sheet; PMC — WP Pathophysiology 2025)
- WP is fat-soluble and can be absorbed systemically through burned areas, causing multi-organ damage including liver, heart, and kidney failure. (CDC/NIOSH — White Phosphorus; WikiDoc — Phosphorus Burns)
- WP particles glow in the dark (luminescent) and fluoresce under ultraviolet (Wood's lamp) light — useful for identifying residual particles. (PMC — WP Pathophysiology 2025; Crisis Medicine)
- Smoke from burning WP causes eye irritation and upper respiratory tract irritation due to phosphorus oxides dissolving in moisture to form phosphoric acids. Delayed pulmonary oedema is possible. (WHO White Phosphorus Fact Sheet; CDC/NIOSH)

Sources: WHO White Phosphorus Fact Sheet (2024); PMC — White Phosphorus Munitions: Pathophysiology 2025 (Frontiers in Public Health); PMC — White Phosphorus Civilian Burns 2024; CDC/NIOSH — White Phosphorus Systemic Agent; Burns Journal — Management of WP Burns

6.2 Appearance and Clinical Features

Feature	Description
Burn appearance	Typically severely painful, necrotic, and yellowish in colour. Characteristic garlic-like odour. Commonly full-thickness burns.
Pain	Severely painful — WP burns involve both thermal and chemical injury.
Particle identification	WP particles are luminescent in dark surroundings and fluoresce under UV (Wood's lamp). Residual particles may re-ignite on exposure to air.

Systemic toxicity phases	Phase 1 (0–8 hours): Gastrointestinal symptoms — abdominal pain, nausea, vomiting, diarrhoea; possible cardiovascular collapse and death within 24–48 hours. Phase 2 (8 hours–3 days): Patient may be asymptomatic. Phase 3 (Day 4–8): Multi-organ failure and CNS dysfunction — may result in death.
Cardiac risk	Fatalities from ventricular arrhythmias have occurred even with < 10% TBSA burns, due to acute hypocalcemia (serum Ca^{2}■ < 1.8 mmol/L). (PMC — WP Pathophysiology 2025)
Electrolyte effects	Hypocalcemia and hyperphosphatemia are predictable complications requiring monitoring and treatment.

Sources: PMC — WP Pathophysiology (Frontiers 2025); PMC — WP Civilian Burns 2024; CDC/NIOSH; Burns Journal — Management of WP Burns 2001; ePlasty — A Phosphorus Burn

6.3 First Aid Protocol — White Phosphorus Burns (Conflict / Low-Resource Setting)

RESCUER SAFETY FIRST: White phosphorus can re-ignite during or after initial treatment on exposure to oxygen. Care providers must have immediate access to clean water or medically prepared saline BEFORE starting treatment. Avoid all sources of ignition (naked flames, electrical equipment, smoking). (WHO White Phosphorus Fact Sheet)

STEP 1	**REMOVE from exposure & isolate oxygen** Remove the patient from the phosphorus source. Immediately isolate the burning area from oxygen by immersing in cold water (below 25°C) or applying water-saturated dressings — replace wet dressings every 5–7 minutes. Warm or hot water MUST be avoided as it can liquefy WP (melting point 44°C), increasing surface contact and exacerbating tissue damage. (WHO White Phosphorus Fact Sheet; PMC — WP Pathophysiology 2025)
STEP 2	**REMOVE contaminated clothing** Carefully remove all contaminated clothing and accessories. Contaminated clothing can ignite or re-ignite — place contaminated items in a water-filled, closable container. (WHO White Phosphorus Fact Sheet; CDC/NIOSH)
STEP 3	**PARTICLE REMOVAL — mechanical debridement** This is a critical step. Partially embedded WP particles must be continuously flushed with water and removed with whatever tools are available (forceps, tweezers, pliers). Remove quickly but gently. Firm or deeply embedded particles that cannot be removed must be covered with saline-soaked dressings kept continuously wet until the patient reaches a medical facility. High-pressure saline from a syringe can dislodge tenacious particles while maintaining cold temperature. (PMC — WP Pathophysiology 2025; WikiDoc — Phosphorus Burns)
STEP 4	**PARTICLE VISUALIZATION** Use ultraviolet light (Wood's lamp) to identify residual WP particles — phosphorus fluoresces under UV light. This is the preferred and safer method. NOTE on copper sulfate: Historically used to visualize particles by turning them black, however a Cochrane Systematic Review (PMC — Interventions for Phosphorus Burns) found no evidence of benefit from copper sulfate and evidence of serious harm (copper absorption causing haemolysis, hepatic necrosis, renal failure, multi-organ failure). Copper sulfate is therefore discouraged. (PMC — Cochrane WP Burns 2020; Crisis Medicine; ePlasty)

STEP 5	**IRRIGATE and COVER**	After particle removal, continuously irrigate with cold saline to ensure complete decontamination. Cover with moist, cool saline-soaked gauze to prevent re-exposure to air and re-ignition during transport. If maintaining gauze moisture is not feasible, hydrogel dressings may be used. Do NOT use soap or alcohol-based solutions during cleaning. (PMC — WP Pathophysiology 2025; Burns Journal — Management of WP Burns)
STEP 6	**SYSTEMIC MONITORING**	Monitor continuously for cardiac arrhythmia — ECG monitoring mandatory. Monitor serum electrolytes hourly for first 12 hours, especially calcium and phosphorus. Correct hypocalcemia with intravenous calcium gluconate (adult and pediatric dose: 0.1–0.2 mL/kg up to 10 mL/dose of a 10% solution). Fluid resuscitation should be adjusted according to urine output. (Burns Journal — Management of WP Burns; ePlasty — A Phosphorus Burn; Crisis Medicine)
STEP 7	**REFERRAL**	WP burns are medical emergencies requiring specialist burn care. Early surgical debridement is often necessary and may need to be repeated until all phosphorus particles are removed. Full-thickness WP burns are necrotic and require excision and skin grafting. There is no antidote for phosphorus toxicity — complete debridement and supportive measures are the key management principles. (Burns Journal; ePlasty; PMC — WP Pathophysiology 2025)

DO NOT apply ice, warm water, or hot water to WP burns.
DO NOT use copper sulfate — associated with serious systemic toxicity (Cochrane Review).
DO NOT use occlusive hydrocolloidal dressings if all WP particles have not been removed.
DO NOT attempt to neutralize WP — water irrigation and mechanical removal are first-line.
DO NOT expose patient to naked flames, electrical equipment, or smoking during treatment.

Sources: WHO White Phosphorus Fact Sheet (2024); PMC — WP Pathophysiology (Frontiers 2025); PMC — Cochrane Interventions for Phosphorus Burns 2020; PMC — WP Civilian Burns 2024; Burns Journal — Management of WP Burns 2001; CDC/NIOSH; WikiDoc — Phosphorus Burns; Crisis Medicine; ePlasty — A Phosphorus Burn

7. OTHER BURN TYPES ENCOUNTERED IN CLINICAL PRACTICE

Beyond the four depth categories and chemical/electrical/WP burns above, the following burn types are recognised in clinical literature and may be encountered in conflict, low-resource, or civilian settings (WHO Fact Sheet; StatPearls; CDC/NIOSH; DermNet NZ):

Radiation Burns

Radiation burns result from protracted exposure to ultraviolet (UV) light (sun, tanning booths, arc welding) or from ionizing radiation (radiation therapy, X-rays, radioactive fallout). Sun exposure is the most common cause of radiation burns and the most common cause of superficial burns overall. (Wikipedia — Burn; DermNet NZ)

- Sunburn is the most common form — treated as a superficial burn.
- Ionizing radiation: hair loss after 3 Gy; redness after 10 Gy; wet skin peeling after 20 Gy; necrosis after 30 Gy. Redness may not appear until some time after exposure. (Wikipedia — Burn)
- Radiation burns are treated the same as other burns of equivalent depth. (Wikipedia — Burn)
- In conflict settings, radiation burns may occur from improvised radiological devices — treat the burn and refer for specialist assessment of systemic radiation exposure.

Sources: Wikipedia — Burn; DermNet NZ — Thermal Burns

Inhalation Injury

Inhalation injury occurs when smoke, hot air, toxic gases, or particles are inhaled. It significantly increases mortality from burn injuries and is a red flag criterion for escalation of care. (MSF Clinical Guidelines; WHO Burn Management)

- Signs: dyspnoea with chest wall indrawing, bronchospasm, soot in nares or mouth, productive cough with carbonaceous sputum, hoarseness, stridor.
- Ensure airway is patent. Administer high-flow oxygen even when SpO_2 appears normal. (MSF Clinical Guidelines)
- In structural fires: consider carbon monoxide poisoning (headache, dizziness with fire-related burns) and cyanide poisoning. (StatPearls — Burn Evaluation; DermNet NZ)
- Inhalation injury consistently requires more fluid — increase fluid resuscitation volumes by 50%. (MSF Clinical Guidelines)
- Burning WP produces dense white smoke of phosphorus pentoxide — irritant to eyes and respiratory mucosa, with risk of delayed pulmonary oedema. (WHO White Phosphorus Fact Sheet)

Sources: MSF Clinical Guidelines; WHO Burn Management; StatPearls — Burn Evaluation; WHO White Phosphorus Fact Sheet

Scald Burns

Scald burns from hot or boiling liquids are among the most common causes of burns, particularly in children. Burns from hot or molten liquids typically produce partial-thickness or full-thickness injuries depending on the temperature, viscosity, and duration of contact. (DermNet NZ — Thermal Burns; WHO Fact Sheet on Burns)

- Steam scalds and cooking oil/fat scalds tend to be deeper than water scalds due to higher temperatures and adhesion to the skin.
- Apply same initial cooling with cool running water for 20 minutes.
- In conflict and low-resource settings, scald burns are a common cause of domestic injury, particularly among women and children cooking over open flames.
- Management follows standard burn depth classification (Section 2).

Sources: DermNet NZ — Thermal Burns; WHO Fact Sheet on Burns

Friction Burns

Friction burns occur when skin is abraded rapidly against a surface, combining both a thermal and mechanical injury. Common in road traffic accidents and blast injuries. (Wikipedia — Burn; StatPearls — Burn Evaluation)

- Appearance: abraded, reddened skin. May be superficial or partial-thickness.
- Clean carefully to remove embedded debris — risk of infection and tattooing if debris is left in wound.
- Treat by depth classification (Section 2) after wound cleaning.
- In blast and conflict settings, friction burns may coexist with shrapnel wounds and require careful wound exploration.

Sources: Wikipedia — Burn; StatPearls — Burn Evaluation

Tar and Bitumen Burns

Tar and bitumen burns occur in industrial or road construction settings. Tar adheres strongly to skin and, if hot, causes deep partial-thickness or full-thickness burns. (Emergency management literature; burn care references)

- Do NOT attempt to peel off adherent tar — this will remove the underlying skin and worsen injury.
- Cool with water to solidify the tar and reduce ongoing thermal injury.
- Tar is removed progressively at dressing changes using mineral oil, petroleum jelly, or lipid-based solvents, which dissolve the tar without damaging the wound.
- Treat underlying burn by depth classification once tar is removed.

Sources: Burn care references; Emergency management literature

8. ESTIMATING BURN SIZE (% TOTAL BODY SURFACE AREA — TBSA)

The size of a burn is measured as a percentage of the total body surface area (TBSA) affected by partial-thickness or full-thickness burns. First-degree (superficial) burns that are only red and not blistering are NOT included and do not influence fluid resuscitation. (MSD Manual; WHO Burn Management)

Rule of Nines — Adults (WHO Burn Management; DermNet NZ; Wikipedia — Parkland Formula):

Body Region	% TBSA
Head and neck	9%
Each upper limb (arm)	9% each
Anterior trunk (chest + abdomen)	18%
Posterior trunk (back)	18%
Each lower limb (leg)	18% each
Perineum / genitalia	1%
Palm of hand (including fingers)	Approximately 1%

Note: The Rule of Nines is modified for children since their heads and lower extremities represent different proportions. The Lund-Browder table provides more accurate estimates for children. (MSF Clinical Guidelines; WHO Burn Management; StatPearls — Burn Fluid Resuscitation)

9. RED FLAGS — CRITERIA FOR ESCALATION TO SPECIALIST CARE

A burn should be considered severe and requiring urgent referral / escalation if any of the following are present (WHO Burn Management; MSF Clinical Guidelines; MSD Manual; WHO Mass Casualty Burns Standards 2024):

Burns involving special areas	Face, ears, eyes, hands, feet, perineum/genitalia, or over a major joint.
Burn size	Adult: greater than 15% TBSA. Pediatric: greater than 10% TBSA.
Full-thickness burns	Any full-thickness burn — requires specialist care and cannot heal without surgery.
Inhalation injury	Dyspnoea, wheezing, soot in nares or mouth, carbonaceous sputum, hoarseness. Ensure airway is patent; high-flow oxygen.
Chemical or electrical burns	Any chemical or electrical burn — treat as high risk regardless of external appearance.
White phosphorus burns	All WP burns are high-risk due to re-ignition potential, systemic toxicity, and cardiac complications.
Circumferential burns	Burns encircling a limb, neck, or chest — risk of compartment syndrome or airway compromise.
Associated trauma / systemic	Shock, hypotension, altered consciousness, or significant associated trauma.
Vulnerable patient groups	Children under 3 years, elderly over 60 years, or patients with significant comorbidities.

10. CORE FIRST AID PROTOCOL (Low-Resource / Conflict Setting)

The following sequential steps apply to all thermal burn types in low-resource or conflict settings (WHO Burn Management; WHO Fact Sheet on Burns; MSF Clinical Guidelines):

STEP 1	**STOP the burning**	Extinguish flames (roll on ground, blanket, or water). Remove clothing. Stop all heat exposure. Ensure your own safety first. For WP burns: smother with water or wet cloth to exclude oxygen.
STEP 2	**COOL the burn**	Cool running water for approximately 20 minutes for burns < 10% TBSA. Do NOT use ice directly. Do NOT apply toothpaste, butter, oil, or home remedies. In large burns (> 10% TBSA) or cold environments, be cautious of hypothermia — especially in children.
STEP 3	**REMOVE constricting items**	Remove rings, watches, bracelets, and tight clothing around burned area early, before swelling.
STEP 4	**COVER the wound**	Wrap with a clean cloth or sheet or non-adherent sterile dressing. Keep patient warm.
STEP 5	**ASSESS severity and ESCALATE**	Reassess burn depth, size, and location. Apply red flag criteria (Section 9). Transport to nearest appropriate facility. The first 6 hours are critical for severe burns.

• Do NOT apply ice directly. • Do NOT apply butter, oil, toothpaste, egg, or any home remedies. • Do NOT attempt to neutralize chemical burns — water irrigation only. • Do NOT remove clothing adherent to the burn. • Do NOT burst blisters. • Do NOT use alcohol-based solutions on the wound. • Do NOT use copper sulfate on WP burns — associated with serious systemic toxicity.

11. SIGNS OF BURN WOUND INFECTION

Initially, burns are sterile. Infection is one of the most frequent and serious complications. Monitor all burns for the following (WHO Burn Management; DermNet NZ):

Burn Wound Cellulitis:

- Progressive redness, swelling, and pain in uninjured skin surrounding the wound.
- Usually appears within the first few days. Caused by Streptococcus pyogenes — responds to penicillin.

Invasive Burn Wound Infection:

- Rapid spread of bacteria from burn eschar into underlying healthy tissues.
- Signs: change in wound colour, new drainage, foul or sickly sweet odour.
- Common causative organisms: Pseudomonas and other gram-negative bacteria.
- Life-threatening — usually requires combined surgical and antibiotic treatment.
- Fever is not a reliable sign of infection — cellulitis in surrounding tissue is a better indicator.

Sources: WHO Burn Management; DermNet NZ; MSF Clinical Guidelines

12. REFERENCES

All content in this guideline is extracted exclusively from the following published sources. No information has been added beyond what is stated in these references.

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[2] **WHO Fact Sheet on Burns**
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[3] **WHO White Phosphorus Fact Sheet (2024)**
World Health Organization. White Phosphorus. Fact Sheet. January 2024. Available at: https://www.who.int/news-room/fact-sheets/detail/white-phosphorus

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